

Ford-Fulkerson algorithm

Theorem: For any network, the value of the maximum flow is equal to the capacity of the minimum cut.

- States that the maximum flow through any network from a given source to a given sink is exactly equal to the minimum sum of a cut.
- This theorem can be verified using the **Ford-Fulkerson algorithm**.
- This algorithm finds the maximum flow of a network or graph.

the ford-fulkerson method (1962)

- is based on 3 bright ideas:
 - **residual flow networks**
the graph that shows where extra capacity might be found
 - **augmenting paths**
paths along which extra capacity is possible
 - **cuts**
used to characterize the flow upper bounds in a network
- the basic method is iterative,
starting from a network with 0 flow:

FORD-FULKERSON-METHOD(G, s, t)

```
1  initialize flow  $f$  to 0
2  while there exists an augmenting path  $p$ 
3      do augment flow  $f$  along  $p$ 
4  return  $f$ 
```

bright idea 1: residual networks

- the **residual network** is a graph with the same vertices as the flow network, yet
- its edges capture the surplus capacity, that is the amount of additional flow we can send through before saturating the edge.

so:

residual capacity $c_f(u,v) = c(u,v) - f(u,v)$

and

$c_f(v,u) = c(v,u) - f(v,u)$

$E_f = \{ (u,v) \in V \times V : c_f(u,v) > 0 \}$

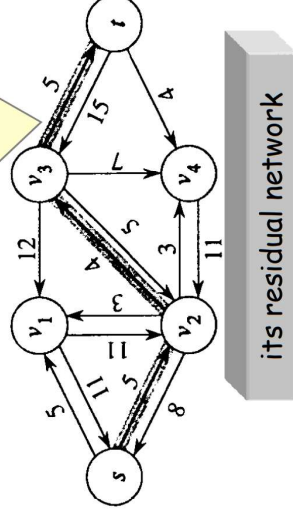
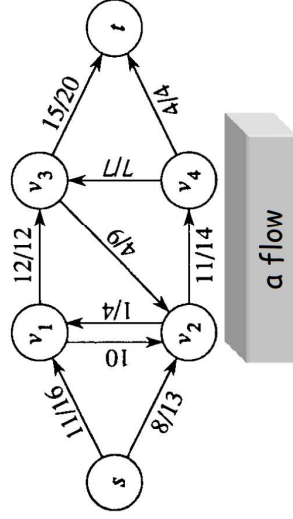
capacity

flow

we maintain capacity in each direction!

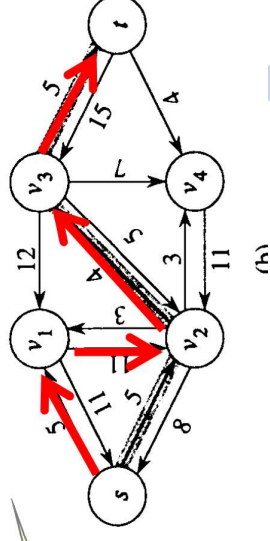
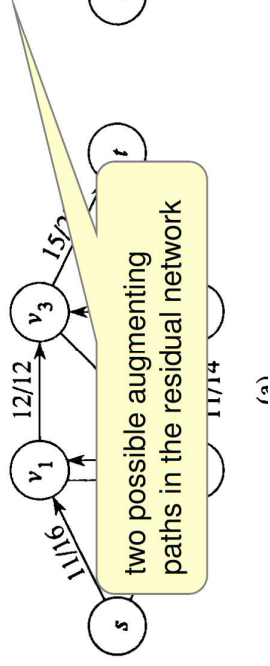
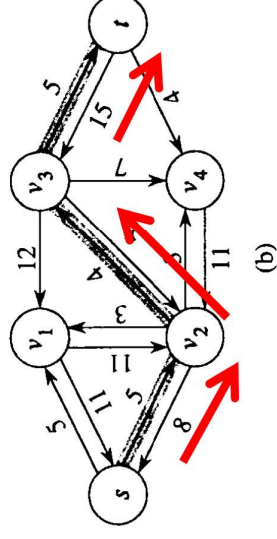
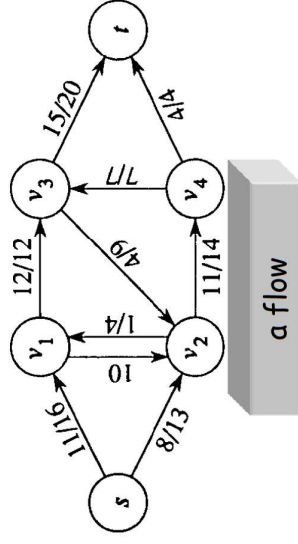
only include edges with a surplus!

5 more megawatts could be sent from v_3 to t , and we could take away 15 before we hit 0



bright idea 2: augmenting paths

- any arbitrary simple path from source to sink that we can find in the residual network shows a way to increase the flow in our network!
- this **augmenting path** is a simple path from s to t in the residual network.

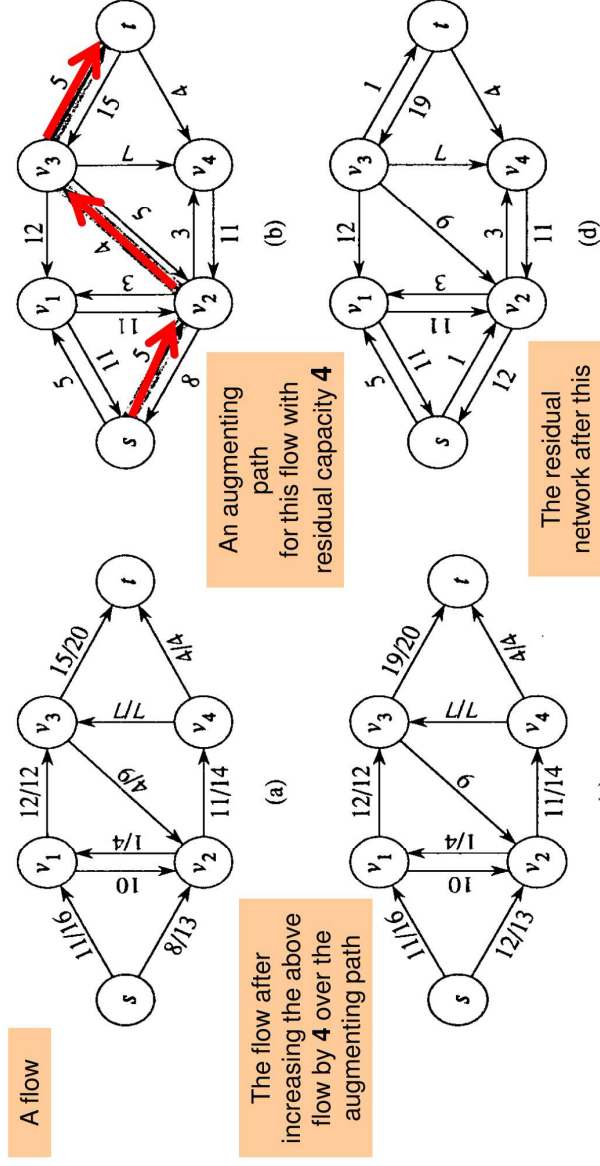


the residual capacity

- the **residual capacity** $c_f(p)$ of an augmenting path p is the maximum additional flow we can allow along the augmenting path, so:

$$- c_f(p) = \min \{c_f(u,v) : (u,v) \text{ is on } p\}$$

The weakest link sets the residual capacity!



ford-fulkerson iterative method

FORD-FULKERSON(G, s, t)

1 **for** each edge $(u, v) \in E[G]$

2 **do** $f[u, v] \leftarrow 0$

3 $f[v, u] \leftarrow 0$

4 **while** there exists a path p from s to t in the residual network G_f

5 **do** $c_f(p) \leftarrow \min \{c_f(u, v) : (u, v) \text{ is in } p\}$

6 **for** each edge (u, v) in p

7 **do** $f[u, v] \leftarrow f[u, v] + c_f(p)$

8 $f[v, u] \leftarrow -f[v, u]$

Start by setting all flows to 0

Pick some augmenting path p which is a path in the residual network

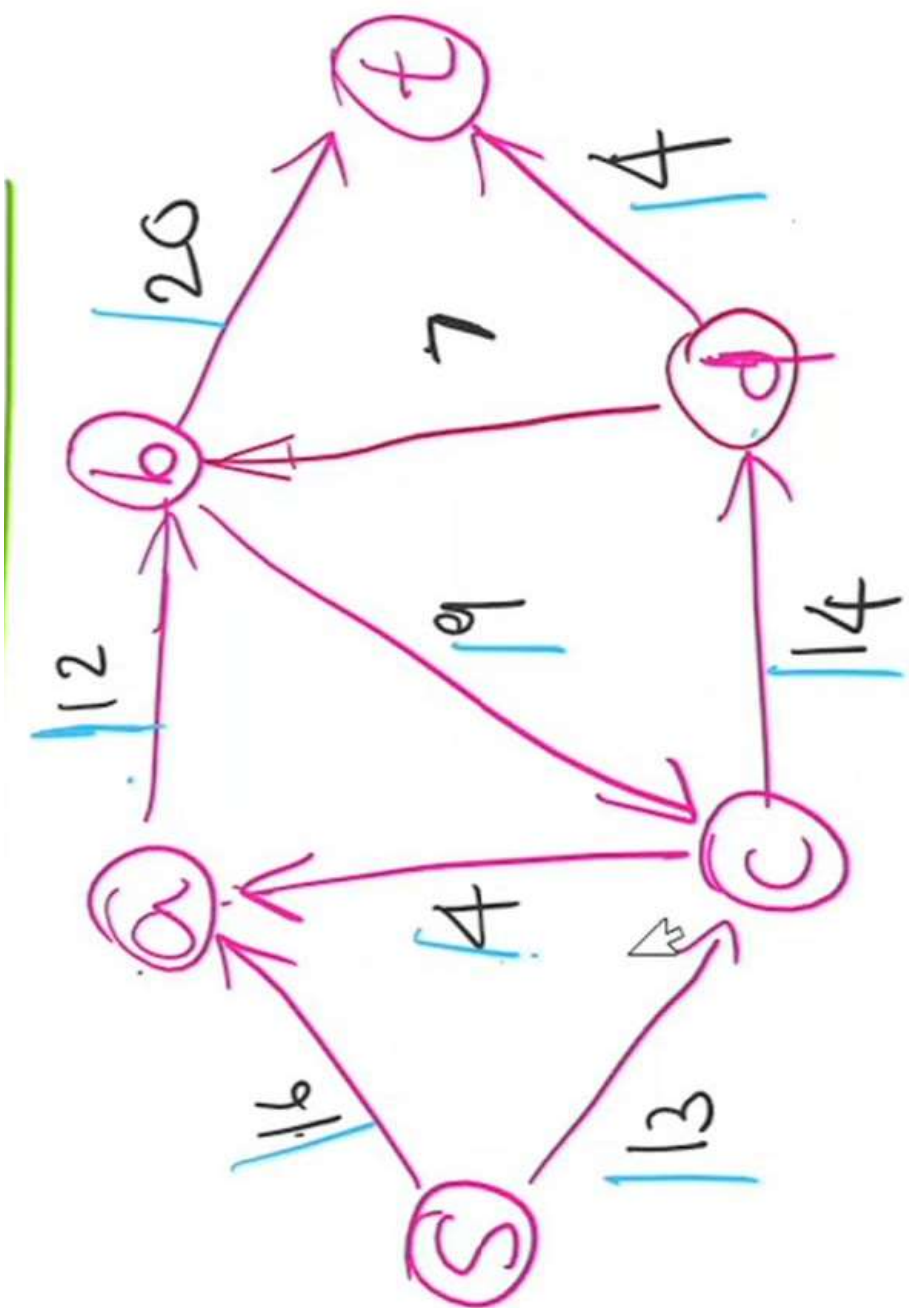
Determine its residual capacity

And increase the flow over the augmenting path by the residual capacity

- the iteration stops when no augmenting path can be found.
- we will see that this results in a maximum flow.

Max flow min cut theorem

- If f is a flow in a flow network G with source s and sink t then the following are equivalent.
 1. f is a max-flow in G
 2. The residual network G_f contains no augmenting path
 3. $|f| = C(S, T)$ for some cut (S, T)
- This acts as proof for Ford Fulkerson algorithm
- To get the cut specified in (3) above (the min-cut) construct S with all vertices that are reachable from the source vertex s in the residual network G_f and T with the remaining vertices.
- Alternatively $S = \{v \in V \mid \text{there exist a path from } s \text{ to } v \text{ in } G_f\}$
- and $T = V - S$
- Thus, (S, T) is as $S - T$ cut since $s \in S$ and $t \in T$



Important observations

- For each edge $(u, v) \in (S, T)$, we have $f(u, v) = C(u, v)$
- For each edge $(v, u) \in (T, S)$, we have $f(u, v) = 0$